CITY UNIVERSITY

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Declaration:

I certify the content of the assignment to be my own and original work, that all sources have been accurately reported and acknowledged and that this document has not been previously submitted in its entirely or in part to any educational establishment

Signature: ______________________________

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Self-Reflective Report

1. Introduction

In this report, I will be detailing my learning experiences from the BIT1123 Object Oriented Programming course. In the tutorials, I was able to learn java programming and the basic concepts of the Object Oriented Programming (OOP).

I finished a total of 10 tutorials. Every tutorial made me learn some new Java concepts and helped me to become better programmer. I began simple classes and objects and later, inheritance, encapsulation, abstraction, file handling and a simple GUI project.

I also got to know how to use Github, VS Code and Command Prompt to arrange and manage my programming work.

2. Knowledge Gained from the Tutorials

Tutorial 1

In Tutorial 1 I had a lesson in the basic structure of a Java program. I developed a class called Student, which has attributes like name, age and GPA. I also gained knowledge of Constructor and Methods.

I learned how to make an object and how to invoke methods from the main program.

Tutorial 2

In Tutorial 2, I built upon my Student class. I gained more knowledge of class design and organization of the class.

I also applied the private attributes and public methods. This gave me an idea of how to think about encapsulation.

Tutorial 3–4

In tutorial 3 and 4 I was introduced to relation between classes. I have worked with Person, Student and Lecturer.

I was introduced to Inheritance and Method Overriding. I also got to know how a child can be incorporated into different class types. This was useful for me to get more understanding about polymorphism.

Tutorial 5

I learned more about encapsulation in Tutorial 5. Used private properties having getters and setters.

For instance, I set and got students' names by calling setName and getName and also set and get their CGPA by calling setCGPA and getCGPA.

It helped me to understand the way I can control the data inside a class.

Tutorial 6

I used Employee and Lecturer classes in Tutorial 6. The Lecturer class was extended from the Employee class.

I got to know about inheritance and how it can be used to reuse code. I also just learned that you can use super() to invoke the parent class constructor.

Tutorial 7

In Tutorial 7 we worked with a class Appliance and two classes, WashingMachine, Refrigerator. I got to know about abstract classes and abstract methods. I also did inheritance and method overriding.

This tutorial gave me some insights into abstraction & polymorphism.

Tutorial 8–9

In Tutorial 8 and 9, I used a simple task program.

I got to know how to use ArrayList to store tasks and Scanner to obtain user input. Also learned how to save data to a file and read it back in.

I utilized FileWriter, BufferedWriter, FileReader and BufferedReader. I also learned how to use try-catch to deal with file errors.

Tutorial 10

I found Tutorial 10 the most challenging tutorial. Made a tiny little GUI quiz program called Quiz Battle.

Learned how to use JFrame, JLabel, and JButton. I also learned about how to use ActionListener so that when the user clicks on the buttons they work.

This tutorial was very helpful to me in linking together various concepts in java in one small program.

3. Challenges Encountered

I should say the most difficult one for me was Tutorial 10.

It was tough because I had to work with both GitHub and VS Code and Command Prompt. Initially, I wasn't feeling very comfortable with Command Prompt and Git commands.

For example, I had some problems when using commands such as git clone and cd to open my project folder.

Another difficulty was grasped some of the concepts of OOP such as inheritance, abstraction and polymorphism. It wasn't so easy for me when I first started.

4. How I Overcame the Challenges

I sought the assistance of my friends in solving my problems. The tutorials and programming homework problems were difficult at times, but my friends Muhammad Fadl, Bawazir Bandar, Ali Ishaq and Ali Sharif were of great assistance.

Using YouTube, I also gained an understanding of some things and technical steps. Your Fix Guide was one of the channels that I used. Here is a tutorial I found on YouTube that I followed to learn how to make and organize my README file for GitHub.

I also repeated the commands and code until I felt that I had a better understanding of it.

This practice and the help helped me feel more comfortable in GitHub, VS Code, Command Prompt and Java Programming.

5. Improvements in My Java Programming Skills

These tutorials helped me grow in java programming skills.

Beginning, I used only simple classes, objects, constructors and methods. Later I was introduced to private variables, getters and setters, inheritance, abstract classes and method overriding.

I also got to learn how to manipulate files, user input, lists and basic GUI based applications.

Now I'm more at ease reading the Java code and what it is doing.

6. Knowledge of the concepts of Object Oriented Programming

The tutorials were useful to me in grasping the core OOP ideas.

Discovered that a class can be used to define objects and the data and methods within them. I found out that an object is an instance of a class. I also learned about encapsulation, which is to keep data private and access through methods.

I have become familiar with the concept of inheritance, in which the child class can inherit properties from the parent class.

I also got to know the concept of polymorphism, which is the ability to have the same method name used differently in various classes.

Lastly, I was introduced to abstraction, particularly the abstraction of an Appliance in Tutorial 7.

These ideas aided me in comprehending ways that Java programs can be organized in a better way.

7. Future Learning Plans

I want to further develop my java skills in the future.

I want to practice build larger Java programs and get a better understanding of GUI (graphical user interface) apps. In addition, I would like to gain more knowledge and learn the application of OOP principles in actual projects.

I would also like to get more comfortable with Git and GitHub since they are a way to manage programming projects.

I want to keep practicing java and have more confidence in my ability to program without assistance.

8. Conclusion

The BIT1123 tutorials provided me with useful experience in programming in java and object-oriented programming.

I began learning basic classes and objects, and as I went on to learn ever more complex concepts, such as encapsulation and inheritance, to abstraction and polymorphism, and finally to file handling and to writing GUI programs.

There were some problems with it, particularly in Tutorial 10, but I managed to overcome them and got help from friends and YouTube tutorials.

This course enhanced my java skills and helped me to understand the object-oriented approach. I think these skills will aid me in my future Programming courses and projects.

9. GitHub Repository URL

GitHub Repository:

https://github.com/Ali-Yousef-Jalal/Ali-Yousef-Jalal-Abdo-202401010131-Java-programming.git